Project Proposal — Edge AI Recyclables Classification Prototype

Theme: Pioneering Tomorrow’s AI Innovations

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Course: AI Future Directions Assignment

1. Project Title

Lightweight Edge AI Model for Real-Time Recyclable Waste Classification Using TensorFlow Lite

2. Background & Motivation

Waste management systems increasingly rely on AI to automate sorting and improve recycling efficiency. However, traditional cloud-based models introduce latency, require stable connectivity, and raise privacy concerns when transmitting camera data. Edge AI solves these limitations by running inference directly on devices such as Raspberry Pi or microcontrollers, enabling real-time decision-making and enhanced data privacy.

This project aims to develop a lightweight image classification model capable of identifying common recyclable materials with low latency and minimal computational cost.

3. Project Goal

Design, train, optimize, and deploy a compact Deep Learning model that performs real-time classification of recyclable items on an Edge AI device.

4. Objectives

- Train a lightweight CNN or MobileNetV2-based model.
- Optimize using TensorFlow Lite.
- Deploy and test inference.
- Measure accuracy, size, and latency.
- Explain Edge AI benefits.

5. Tools & Technologies

TensorFlow, TensorFlow Lite, Google Colab, Raspberry Pi, Python, OpenCV.

6. Dataset

TrashNet or Kaggle recyclable items dataset.

7. Methodology

Data preprocessing, model design, training, TFLite conversion, evaluation, edge deployment simulation.

8. Expected Deliverables

Notebook, TFLite model, inference script, report, reflection.

9. Expected Impact

Demonstrates how Edge AI enables real-time, privacy-preserving, low-power intelligent systems to support smart waste management.